

The $^{27}\text{Al}(n,\gamma)$ capture rate on the He-3 capsule: an analytic cross-check of Geant4

Claude (Fable 5) for Dylan
MX17 / nTOF EAR2

2026-07-23

Task: `.claude/al_gamma_yield_check/HANDOFF.md` — is the Geant4 Al-capture background rate right, and why did a back-of-envelope estimate look low?

The question

Geant4 says the Al pressure vessel of the He-3 capsule glows:

- ▶ $5.95 \times 10^6 \gamma \rightarrow e^+ e^-$ pairs/day produced from the 7.72 MeV $\text{Al}(n, \gamma)$ line — the dominant $e^+ e^-$ background to the X17 search.

A hand calculation (13 g Al, single-pass slab, beam below 1 eV) gave a γ yield that *looked low*.

This cross-check compares at the cleanest level: *radiative captures in the Al*, before any pair-conversion folding.

Comparison quantity

EventTree, 20 files $\times 10^7$ thermal neutrons ($E \in [1 \text{ meV}, 2 \text{ eV}]$):

`capture_vol == He3Cap_Al`

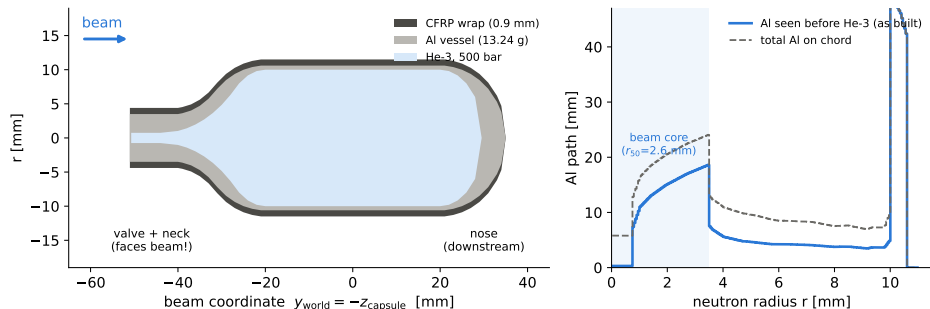
`capture_proc == nCapture`

1,561,193 captures / 2×10^8 n

$= 7.81 \times 10^{-3}$ per neutron

($\pm 0.08\%$)

First finding: the capsule faces the beam valve-first



The placement rotation (`rotateX(-90°)`) maps local $z \rightarrow -y_{\text{world}}$: the **valve** (local $z = +51 \text{ mm}$) sits at $y = -51 \text{ mm}$, *upstream*, facing the gun.

The beam core ($r_{50} = 2.6 \text{ mm}$) bores through the nearly solid Al valve stem + shoulder: **10–20 mm of Al**, not 5.5 mm of nose, not 0.6 mm of wall.

How we know the orientation (it's in the data)

Geant4 records the capture position per event ($\text{cap_x}, y, z$). Where the Al captures actually happen:

capsule region	captures
valve stem ($z \in [40, 51]$)	68.3%
shoulder taper ($z \in [21, 40]$)	28.5%
barrel wall ($ z < 21$)	3.0%
nose dome ($z < -21$)	0.2%

96.6% of captures at $r < 10$ mm — the valve/neck, not the barrel wall.

If the nose faced the beam

the nose would capture first and the valve would hide behind 17 cm^3 of black He-3. Instead the nose is *empty* and the valve *saturated*: the valve faces the beam.

Why it matters

He-3 at 500 bar is optically thick ($\lambda_{\text{abs}} \approx 25 \mu\text{m}$ at 25 meV): whatever Al lies *behind* first gas contact is invisible. Orientation alone changes the capture rate by $\times 1.8$.

Ingredients of the analytic model

Beam (exactly as the sim samples it)

- ▶ energy: EAR2 flux histogram,
 $\Phi = 4.29 \times 10^6$ n/pulse in [1 meV, 2 eV]
(3.99×10^6 below 1 eV)
- ▶ radius: Lambda2D profile row nearest in
 $\log E$ ($r_{50} = 2.6$ mm, $r_{90} = 13$ mm; 86%
inside the capsule face)
- ▶ 1.929×10^4 pulses/day

Geometry: exact STEP polycones,
mirrored from `DetectorConstruction.cc`;
reproduces $V_{\text{Al}} = 4.907 \text{ cm}^3$,
 $V_{\text{gas}} = 17.003 \text{ cm}^3$ to 4 digits.

Nuclear data (ENDF / IAEA EGAF)

- ▶ $^{27}\text{Al}(n,\gamma)$: $\sigma_0 = 0.231$ b at 25.3 meV,
 $1/\nu$ below the 5.9 keV resonance
- ▶ cross-check: the ^{28}Al β -decay line
(1778.9 keV, ~ 1 per capture) has
 $\sigma_\gamma = 0.232(4)$ b ✓
- ▶ **7724 keV line:** $\sigma_\gamma = 0.0493(15)$ b
 $\Rightarrow$ **21.3% of captures** (not 1 per
capture!)
- ▶ $^3\text{He}(n,p)t$: ENDF (5333 b at
25.3 meV, $1/\nu$)
- ▶ Al elastic ≈ 1.4 b ($6\times$ capture:
ignored in single-pass, checked
against Geant4)

Method: single-pass straight-line attenuation

$$R = \sum_{E \text{ bins}} \Phi(E) \int f(r|E) dr \underbrace{\int_{\text{path}} dz \Sigma_{\gamma}^{\text{Al}}(E) \kappa_{\text{Al}}(r, z) e^{-\Sigma_{\gamma}^{\text{Al}} t_{\text{Al}}(z) - \Sigma_{\text{abs}}^{\text{He3}} t_{\text{gas}}(z)}}_{\text{capture prob. on the straight chord at radius } r}$$

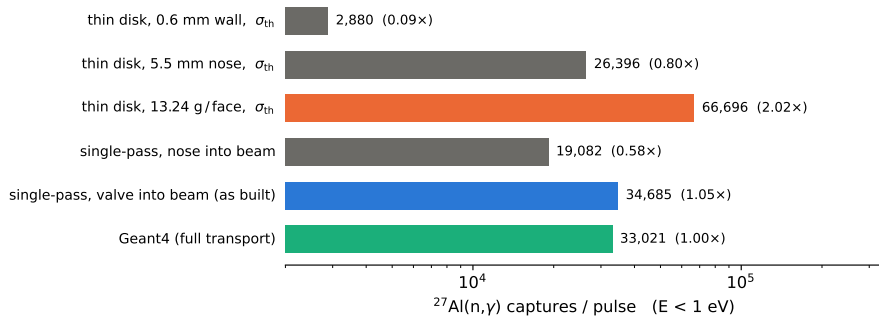
Includes

- ▶ exact thickness-vs-radius profile (0.02 mm marching)
- ▶ He-3 shadowing with the true $\sigma(E)$ (the gas is black below ~ 1 keV)
- ▶ $1/v$ folded over the real flux spectrum, bin by bin
- ▶ the same radial-profile-vs-energy correlation the gun uses

Deliberately ignores (checked a posteriori)

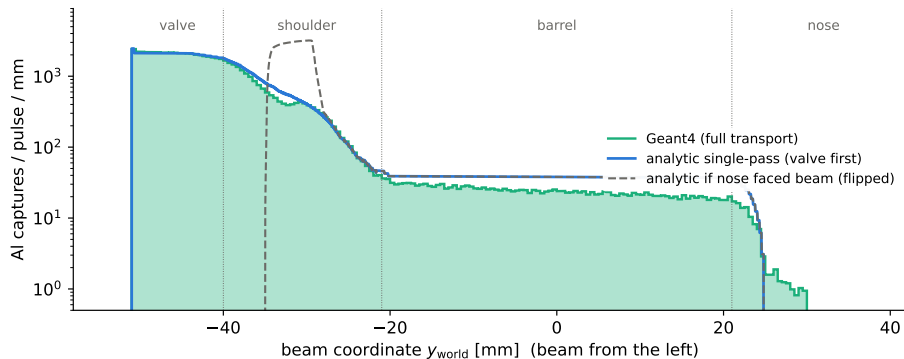
- ▶ elastic scattering in Al/CFRP/air (redirects, $\sim 6:1$ over capture)
- ▶ albedo from the surrounding plastics/LS
- ▶ these partially cancel: out-scatter $-$, re-entry $+$

Result: the ladder from “thin disk” to truth



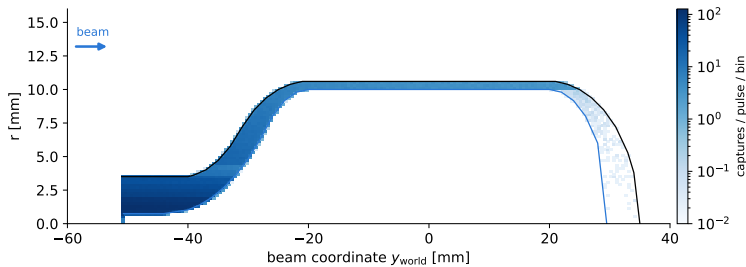
Wall-slab: 11 $\times$ low. Mass-smeared: 2 $\times$ high. Single-pass on the true geometry: within 5% of Geant4.

The capture map: the model predicts *where*, too



Geant4 and the straight-line model agree over three decades along the capsule. The counterfactual flipped mounting (dashed) would capture $1.8\times$ less, concentrated in the nose dome.

Same story in 2D and in energy



Captures hug the upstream Al surfaces (Geant4; outlines: Al black, gas blue). The energy spectrum of capturing neutrons matches shape-wise with a flat ~ 0.95 ratio (backup slide).

What the thin-disk assumptions cost — ranked

effect	factor	comment
which Al the beam sees		the dominant issue
0.6 mm wall slab	×0.09	beam core never sees the wall
5.5 mm nose slab	×0.8	fortuitously close, wrong picture
13.24 g smeared over the face	×2.0	counts shadowed Al (3.7 g behind gas) and ignores where flux is
orientation (valve vs nose first)	×1.8	as-built: valve first, $\langle t_{\text{pre}} \rangle = 9.1 \text{ mm}$
σ_{th} instead of $\langle \sigma \rangle_{\text{flux}}$	×1.45	the $<1 \text{ eV}$ window is <i>above</i> 25 meV on average; sub-thermal $1/v$ gain never compensates
straight line vs full transport	×0.95	scattering out $\approx$ albedo in; He-3 absorbs 77% (G4) vs 91% (straight line)
assuming 1 γ (7.72 MeV)/capture	×4.7	EGAF: 21.3% of captures

The quotable one-line calculation

$$R = \Phi \times n_{\text{Al}} \bar{\sigma}_{\gamma} \times \langle t_{\text{pre}} \rangle = 4.29 \times 10^6 \times (6.02 \times 10^{22} \frac{1}{\text{cm}^3} \times 0.150 \text{ b}) \times 9.1 \text{ mm} \approx 3.5 \times 10^4 \text{ captures/pulse}$$

- ▶ $\bar{\sigma}_{\gamma} = 0.231 \text{ b} \times 0.65$: thermal cross-section folded with the actual [1 meV, 2 eV] spectrum ($1/v$)
- ▶ $\langle t_{\text{pre}} \rangle = 9.1 \text{ mm}$: flux-weighted Al path *before first He-3 contact* (valve-first); the only nontrivial geometry number
- ▶ optically thin ($\Sigma_{\gamma} \langle t \rangle \approx 0.8\%$), so everything factorizes

[1 meV, 2 eV]	per pulse	per day	vs Geant4
one-line / full single-pass	35,180	6.79×10^8	+5%
Geant4	33,508	6.46×10^8	—

From captures to the 7.72 MeV γ (and to pairs)

(Geant4-anchored)	per pulse	per day
Al captures, $E_n < 2$ eV	33,508	6.5×10^8
$\gamma(7.724 \text{ MeV})$, 21.3%	7,151	1.4×10^8
$\gamma \rightarrow e^+ e^-$ anywhere	~ 328	6.4×10^6
(of which in Al)	308	6.0×10^6

Extending to $E_n < 1$ keV adds only **+3.6%** captures (analytic): Φ grows $1.7\times$ but $\sigma \sim 1/v$ dies.

- ▶ pair conversion of the 7.72 MeV line costs another $\sim 2\%$ (from the sim's own conv-pair campaign) — *that* chain, not the capture rate, is where the small numbers come from
- ▶ if a hand calc also used 1 γ /capture and the 0.6 mm wall, errors of $\times 4.7$ (up) and $\times 11$ (down) fight: “looks low” is the expected outcome

Conclusions

1. **The Geant4 number is right.** An independent straight-line calculation on the true geometry reproduces the capture rate to **+5%** and the capture *locations* and *energy spectrum* in detail. Nothing about the 6×10^8 captures/day (1.4×10^8 7.72 MeV γ /day) is anomalous.
2. **Thin-disk is inadequate here** unless “thickness” means the flux-weighted pre-gas Al path (9.1 mm). Wall/nose/mass variants land anywhere from $0.09\times$ to $2\times$.
3. The two knobs that matter beyond geometry: $\langle\sigma\rangle/\sigma_{\text{th}} = 0.65$ (window average) and 21.3% for the 7.724 MeV line.
4. **Action item: confirm the capsule mounting.** The sim as built has the *valve facing the beam*; the handoff (and probably our mental model) assumed nose-first. If the real mounting is nose-first, the Al capture background drops $\times 0.55$ — and the sim placement (`rotateX`) needs a sign fix, re-running the Al-pair background chain.

Part II: from γ s to single-arm coincidences

Alberto's calculation (email 2026-07-23) vs this work, per pulse:

	Alberto	this work (G4)	ratio
Al captures, $E_n < 1$ eV model	72,787	33,021	2.2
	13 g thin disk, $r \approx 1$ cm	full transport	
7.72 MeV γ intensity	26.8 %	21.3 % (EGAF)	1.26
γ_0 produced, $E_n < 1$ eV	19,507	7,033	2.8
γ_0 into detector	878 ($\Omega=4.5$ %)	see next slide	
single-arm coincidences	~ 9 ($\times 1$ %)	199	~ 22

The capture/ γ level **agrees to a understood factor ~ 2** (his thin disk = the “mass-smeared” rung of our ladder; bin-by-bin ratio 2.2–2.4 below 1 eV, same spectral shape). The **$\times 22$ tension** is entirely downstream of the γ yield.

What the sim counts: “legs” (SiPM $\wedge$ plastic per arm)

Per arm: S = max single SiPM-bar edep, P = max plastic-bar edep; leg $\equiv S \geq 0.5 \text{ MIP} \wedge P \geq 0.5 \text{ MIP}$, in-gate ($>1 \text{ ms}$).

legs/pulse (arm-summed)	2.0 cm	2.5 cm
baseline	203.4	168.5
no-Al capsule	4.6	3.3
Al-attributable	198.8	165.2

$\approx$ **50 legs/arm/pulse** — Dylan’s number ✓

Thresholds in energy:

- ▶ SiPM bar (3 mm PVT): $0.5 \text{ MIP} =$ **0.23 MeV**
- ▶ plastic (2.0 cm PVT): $0.5 \text{ MIP} =$ **1.73 MeV** (the 1.78 MeV ^{28}Al decay line is delayed 2.2 min *and* below max-Compton threshold — irrelevant)

Apertures: Alberto's 4.5 % is ONE arm of four

From the capture centroid (valve, $y \simeq -4.5$ cm):

surface	size	distance	$\Omega/4\pi$ per arm	$\times 4$ arms
SiPM wall (3 mm)	40 $\times$ 50 cm	33.2 cm	10.1 %	40.3 %
plastics (2 cm)	40.3 $\times$ 30 cm	$\sim$ 42 cm	4.6 %	18.5 %

Alberto's "solid angle covered = 4.5 %" equals one arm's plastic aperture almost exactly. The coincidence-limiting aperture (plastics, 4 arms) is **$\times 4.1$ larger**; the SiPM wall sees $\times 9$ his solid angle.

Scintillator material is NOT the explanation ($\pm 10\%$)

Compton is electron-density physics; every CH-based plastic is the same to $\pm 10\%$.
Klein–Nishina interaction probabilities:

material	$n_e [10^{23}/\text{cm}^3]$	$P_C(2\text{ cm}, 7.72\text{ MeV})$	$P_C(2\text{ cm}, 3.03\text{ MeV})$
PVT (EJ-200 family; sim)	3.37	4.05 %	7.40 %
polystyrene-based	3.40	4.09 %	7.48 %
PMMA (acrylic)	3.87	4.64 %	8.46 %
polyethylene	3.23	3.89 %	7.11 %

Pair production adds $\sim 0.9\%$ at 7.72 MeV (2 cm). The 3 mm SiPM bar has $P_C = 0.62\%$ at 7.72 MeV — **Alberto's “ $\sim 1\%$ Compton” describes the thin SiPM bar, not the 2 cm plastic (4–5 %, and 7–8 % for the 3–4.7 MeV cascade lines).**

Geant4's answer: ONE electron makes both signals

Two truth runs: every leg classified in 6×10^7 neutron events (2,868 legs, real cascade), and a dedicated γ -**source truth run** (4×10^6 single cascade γ s from real capture vertices, 11,181 legs, with birth volume/process recorded per track):

leg mechanism (combined truth)	fraction
one track crosses bar+plastic	98 %
via hard Compton	80.0 %
via pair production	17.9 %
birth KE (median)	4.5 MeV
same γ , two interactions	0.4 %
two different cascade γ s	1.2 %
crossing track also crossed the MM	52 %

The broken assumption

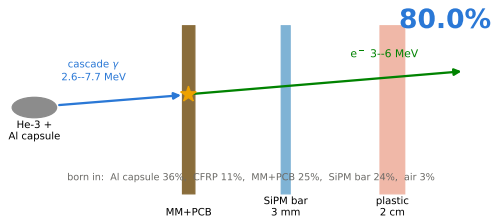
A coincidence does NOT cost $P_{\text{SiPM}} \times P_{\text{plastic}}$ — one interaction anywhere upstream buys both signals: the multi-MeV electron crosses the 3 mm bar (0.23 MeV) and buries itself in the 2 cm plastic (1.73 MeV). Measured response to an emitted 7.72 MeV γ : **0.54 %** (all arms) = 2.9 % per γ through the plastic aperture.

Half the leg electrons are born at the *capsule* and fly ~ 33 cm through the MM — they leave an MM track (veto handle!).

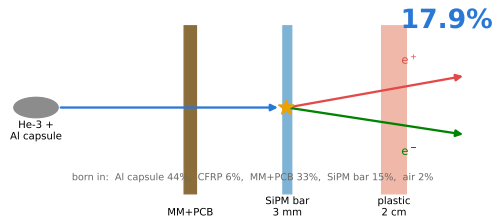
The four ways a leg happens — with measured fractions

How an AI capture γ makes a SiPM $\wedge$ plastic leg — Geant4 truth (topology: neutron run, 1406 legs; origins: γ -source truth run, 11,181 legs)

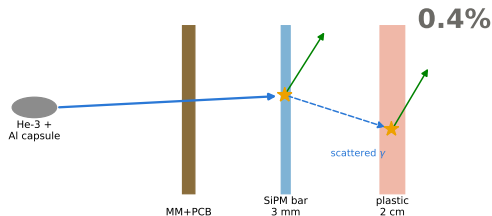
(A) hard Compton, ONE e^- crosses bar + plastic



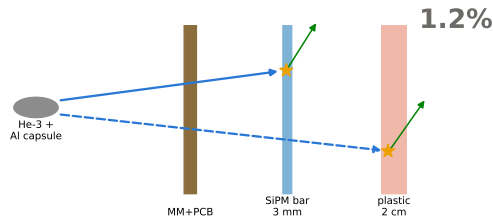
(B) pair production, e^+ / e^- cross bar + plastic



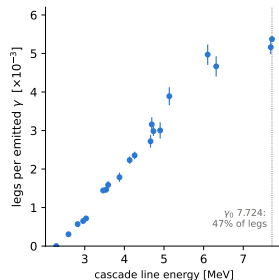
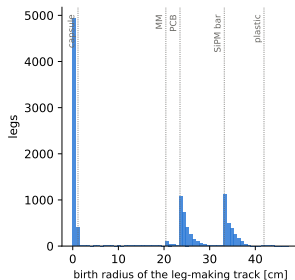
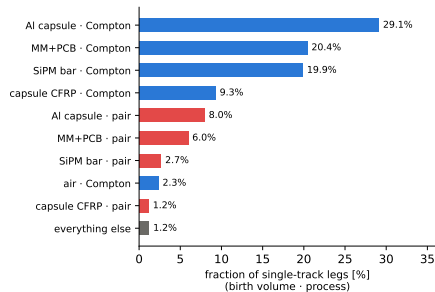
(C) same γ interacts twice (old ``double Compton'')



(D) two different cascade γ s, same arm

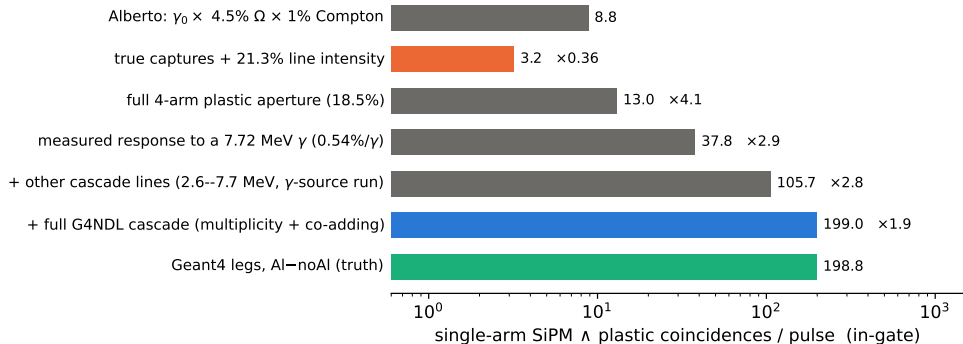


Where the leg-making electron is born, and which line sent it



Left: birth volume $\times$ process (Compton blue, pair red). Middle: birth radius — the capsule itself is the single biggest electron source, then the MM PCB and the SiPM bar; essentially nothing starts in the plastic. Right: legs per emitted γ vs line energy — zero below 2.3 MeV (plastic threshold), rising steeply; the 7.724 MeV line alone makes **47% of all legs**.

The reconciliation ladder: 9 $\rightarrow$ 199 legs/pulse



Every rung is now *measured*: the 7.72- γ response (0.54 %/emitted γ) and the cascade-line sum come from the γ -source truth run; the last rung ($\times 1.9$) is the full G4NDL cascade vs the 20-line discrete list (multiplicity/continuum + within-cascade co-adding). No single error — five compounding assumptions. The scintillator *material* is irrelevant ($\pm 10\%$).

Orientation history: it was never flipped — it was born this way

2026-06-01 `rotateX(-90°)` + beam + Y introduced — capsule still a symmetric cylinder (G4Tubs); orientation meaningless.

2026-06-10 STEP polycone lands (tip $z = -35$, valve $+51$) under the *pre-existing* rotation $\Rightarrow$ **valve-upstream from day one**.

2026-06-11 thermal_note: prose says beam “enters through a 5 mm-thick domed aluminium tip” (the intention), but its own quoted rate — **capture in Al capsule** 8.0×10^{-3} — is the *valve-first* number (nose-first = 4.5×10^{-3}). The sim was already flipped then.

06-30 / 07-14 / 07-15 axis convention, MM spacing, stack flip: none touch `capRot` or the polycone z arrays (verified by diff).

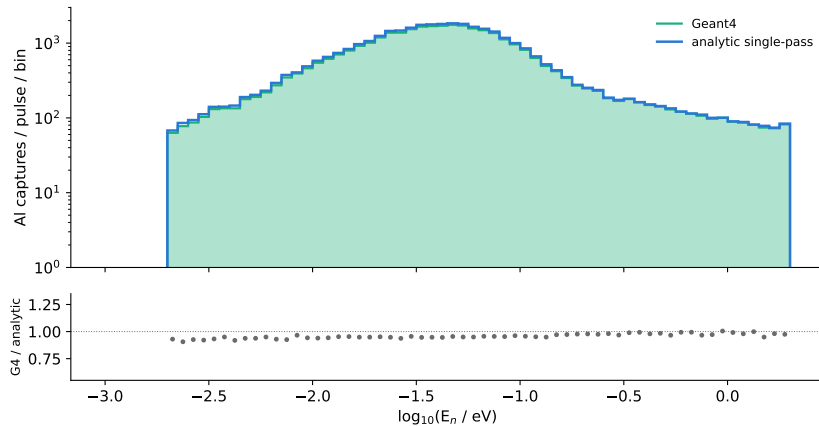
2026-07-22 event display draws $y_{\text{world}} = -z_{\text{local}}$ (correct, valve upstream).

Silver lining: every thermal-era result is internally consistent — all valve-first.

Conclusions — Part II

1. **The sim's ~ 50 legs/arm/pulse (199 summed) is right** — and Alberto's 9/pulse is right *for his assumptions*. The gap: four-arm aperture ($\times 4.1$), the real response to a $7.72\ \gamma$ ($\times 2.9$ — a *single* 3–6 MeV Compton/pair electron crosses both detectors, 80/18 %), the rest of the cascade ($\times 2.8$), and the full-cascade correction ($\times 1.9$). Half the leg electrons are born at the capsule and cross the MM $\Rightarrow$ an MM-track veto could kill $\sim 50\%$ of trigger legs.
2. **The plastic material is not the variable**: every CH-based scintillator is within $\pm 10\%$ (electron-density physics). The thicknesses and the upstream material (MM PCB, container, bar) are what matter.
3. Capture/ γ -yield level: we and Alberto agree once his thin disk is corrected by the known $\times 2.2$ (geometry + shadowing + σ folding) and $26.8\% \rightarrow 21.3\%$ (EGAF).
4. Orientation: valve-first since the STEP capsule existed (2026-06-10); nothing was flipped by the axis change. All thermal-era numbers are internally consistent. **Still open: which way is the *real* capsule mounted?** (nose-first $\Rightarrow$ captures $\times 0.55$, and the capture zone moves 14 cm downstream — leg rates would change too.)

Backup: energy spectrum of capturing neutrons



Backup: γ -source truth run bookkeeping

- ▶ New HitTree branches (`origin_vol/proc/ke`, `ox,oy,oz`): birth truth of every depositing track (`GetLogicalVolumeAtVertex`, `GetCreatorProcess`). In the repo (uncommitted) + built on `lxplus`.
- ▶ γ -source mode: one cascade γ /event, 20 discrete ^{28}Al lines >2 MeV (IAEA intensities, $\Sigma I = 1.13/\text{capture}$), isotropic from 156k real `He3Cap_Al` capture vertices (`make_capture_library.py`, thermal campaign). 4×10^6 events $\rightarrow$ 11,181 legs.
- ▶ **Closure:** γ -source predicts 2.80×10^{-3} legs/ $\gamma \times 1.13 \times 33,508$ captures/pulse = **106/pulse** vs 199 in full neutron mode $\Rightarrow$ the discrete-line list underestimates the true G4NDL cascade response $\times 1.9$ (continuum multiplicity + within-cascade co-adding). Mechanism *fractions* are validated against the neutron run: capsule+air-born 50.3 % vs MM-crossing 51.6 %; e^+ 9.2 vs 7.3 %; birth KE 4.5 MeV vs 3.9 MeV at the bar (≈ 0.5 MeV en-route loss).
- ▶ Old 84/16 slide (2026-07-22): the 16 % pair fraction was right (here 18 %); the “84 % double Compton” *topology* was wrong — true rate 0.4 %.

Backup: caveats & bookkeeping

- ▶ Geant4 side: 20/100 files of `neutrons_thermal_trig_2cm` (analog, no He-3 capture bias; `weight=1`), stat. error 0.08%.
- ▶ “Captures” = terminal `nCapture` of the *primary* track. $^3\text{He}(n,p)t$ is filed under `neutronInelastic` by G4NeutronHP — He-3 absorption is 77.4% of neutrons, not the ~ 2 “He3Gas `nCapture`” entries (radiative branch).
- ▶ $\langle t_{\text{pre}} \rangle$ folds the per-energy radial profiles; quoting it as a single number hides a mild energy dependence (ratio panel stays within $\pm 7\%$).
- ▶ < 1 keV extension is analytic-only (campaign window ends at 2 eV); resonance region (≥ 5.9 keV) not addressed.
- ▶ 7724 keV intensity from IAEA PGAA/EGAF k0 table ($\sigma_\gamma = 0.0493(15)$ b). Geant4’s internal capture-cascade branching was *not* independently unfolded here; for note-level γ yields use EGAF.
- ▶ Air (16 cm), CFRP scattering, thermal-scattering-law effects: all inside the residual -5% .